

When is a hypergeometric function algebraic?¹

Differential Seminar (IHP, Paris)

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¹Joint work (arxiv.org/abs/2308.12855) with Florian Fürnsinn

Definitions

Hypergeometric sequence:

$$u_{n+1} = \frac{A(n)}{B(n)} u_n,$$

where $A(t), B(t) \in \mathbb{Q}[t]$ are polynomials.

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Hypergeometric function:

$${}_pF_q \left[\begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_q \end{matrix} ; x \right] := \sum_{n=0}^{\infty} \frac{(a_1)_n \cdots (a_p)_n}{(b_1)_n \cdots (b_q)_n} \cdot \frac{x^n}{n!},$$

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Hypergeometric functions are generating functions
of **hypergeometric sequences**.

Examples

- The logarithm:

$${}_2F_1\left[\begin{matrix} 1, 1 \\ 2 \end{matrix}; x\right] = -\frac{\log(1-x)}{x} = 1 + \frac{1}{2}x + \frac{1}{3}x^2 + \frac{1}{4}x^3 + \dots \in \mathbb{Q}[[x]]$$

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- Catalan numbers:

$$C_n = \binom{2n}{n} \frac{1}{n+1} \in \mathbb{Z}, \quad \frac{C_{n+1}}{C_n} = \frac{(2n+1)(2n+2)}{(n+1)(n+2)}, \quad \sum_{n \geq 0} C_n x^n = {}_2F_1\left[\begin{matrix} \frac{1}{2}, 1 \\ 2 \end{matrix}; 4x\right]$$

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- Chebychev numbers:

$$C_n = \frac{(30n)!n!}{(15n)!(10n)!(6n)!} \in \mathbb{Z}, \quad \sum_{n \geq 0} C_n x^n = {}_8F_7\left[\begin{matrix} \frac{1}{30}, \frac{7}{30}, \frac{11}{30}, \frac{13}{30}, \frac{17}{30}, \frac{19}{30}, \frac{23}{30}, \frac{29}{30} \\ \frac{1}{5}, \frac{1}{3}, \frac{2}{5}, \frac{1}{2}, \frac{3}{5}, \frac{2}{3}, \frac{4}{5} \end{matrix}; 2^{14} \cdot 3^9 \cdot 5^5 \cdot x\right]$$

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- More algebraic series:

$${}_3F_2\left[\begin{matrix} 1/2, \sqrt{2}+1, -\sqrt{2}+1 \\ \sqrt{2}, -\sqrt{2} \end{matrix}; 4x\right] = \frac{(7x-1)(2x-1)}{(1-4x)^{5/2}} = 1 + x - 6x^2 + \dots \in \mathbb{Z}[[x]]$$

Definitions

A series $f(x) \in \mathbb{Q}[[x]]$ is called **algebraic** if there exists $P(x, y) \in \mathbb{Q}[x, y] \setminus \{0\}$, such that $P(x, f(x)) = 0$.

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A series $f(x) \in \mathbb{Q}[[x]]$ is called **almost integral** if there exist $\alpha, \beta \in \mathbb{Z} \setminus \{0\}$, such that $\beta f(\alpha x) \in \mathbb{Z}[[x]]$.

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Theorem (Eisenstein 1852, Heine 1854)

Any algebraic $f(x) \in \mathbb{Q}[[x]]$ is almost integral.

$\Rightarrow$ Arithmetic proof that $-\log(1-x)/x = 1 + \frac{1}{2}x + \frac{1}{3}x^2 + \frac{1}{4}x^3 + \cdots$ is transcendental.

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Conjecture (weak form of p -curvature conjecture, Grothendieck 1969, Bézivin 1991)

A linear ODE of order n has n linearly independent algebraic solutions if and only if it has n linearly independent almost integral solutions.

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Any hypergeometric function $F(x) \in \mathbb{Q}[[x]]$ is D-finite:

$$x(\theta + a_1) \cdots (\theta + a_p)F(x) = \theta(\theta + b_1 - 1) \cdots (\theta + b_{p-1} - 1)F(x), \quad \left(\theta = x \frac{d}{dx}\right).$$

For example, $F(x) = {}_2F_1\left[\begin{smallmatrix} a, b \\ c \end{smallmatrix}; x\right]$ satisfies

$$x(1-x)F''(x) + [c - (a+b+1)x]F'(x) - abF(x) = 0.$$

Main Question of this Talk

Which hypergeometric functions are algebraic?

$${}_2F_1\left[\begin{matrix} 1, 1 \\ 2 \end{matrix}; x\right]$$

$${}_2F_1\left[\begin{matrix} \frac{1}{2}, 1 \\ 2 \end{matrix}; 4x\right]$$

$${}_3F_2\left[\begin{matrix} 1/2, \sqrt{2}+1, -\sqrt{2}+1 \\ \sqrt{2}, -\sqrt{2} \end{matrix}; 4x\right]$$

$${}_6F_5\left[\begin{matrix} \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 3, 3, 1 \\ \frac{1}{3}, \frac{2}{3}, 4, 2, 2 \end{matrix}; \frac{256}{27}x\right]$$

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$${}_3F_2\left[\begin{matrix} \frac{5}{6}, \frac{1}{2}, 1 \\ 2, \frac{5}{3} \end{matrix}; 16x^2\right]$$

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$${}_2F_1\left[\begin{matrix} 1, 1 \\ 2 \end{matrix}; x\right] = -\frac{\log(1-x)}{x}$$

$${}_2F_1\left[\begin{matrix} \frac{1}{2}, 1 \\ 2 \end{matrix}; 4x\right] = \frac{1-\sqrt{1-4x}}{2x}$$

$${}_3F_2\left[\begin{matrix} 1/2, \sqrt{2}+1, -\sqrt{2}+1 \\ \sqrt{2}, -\sqrt{2} \end{matrix}; 4x\right] = \frac{(7x-1)(2x-1)}{(1-4x)^{5/2}}$$

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Schwarz' Classification (Geometry & Algebra)

- **[Schwarz 1873]**: Classification of all algebraic **Gaussian hypergeometric functions**:

$$F(x) = {}_2F_1([a_1, a_2], [b_1]; x),$$

with rational parameters $a_1, a_2, b_1 \in \mathbb{Q} \setminus \mathbb{Z}$.

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- Write $(\lambda, \mu, \nu) = (1 - b_1, b_1 - a_1 - a_2, a_2 - a_1)$.

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Example

$F(x) = {}_2F_1([-1/2, -1/6], [2/3]; x)$ is algebraic, as $(\lambda, \mu, \nu) = (1/3, 4/3, 1/3)$ and $(-(\nu - 1), \lambda, \mu - 1) = (2/3, 1/3, 1/3)$ is in the list.

No.	λ''	μ''	ν''	$\frac{\text{Inhalt}}{\pi}$	Polyeder
I.	$\frac{1}{2}$	$\frac{1}{2}$	ν	ν	Regelmässige Doppelpyramide
II.	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{6} = A$	Tetraeder
III.	$\frac{2}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3} = 2A$	
IV.	1	1	1	1 = D	

Landau-Errera Criterion (Arithmetic)

[Landau 1904, 1911]: Necessary condition for algebraicity of Gaussian hypergeometric functions (based on Eisenstein's criterion):

Theorem (Landau)

Let $F(x) = {}_2F_1([a_1, a_2], [b_1]; x)$ with $a_1, a_2, b_1, a_1 - b_1, a_2 - b_1 \notin \mathbb{Z}$ and N their common denominator. Then $F(x)$ is almost integral if and only if for all $1 \leq \lambda \leq N$ coprime to N we have

$$\langle \lambda a_1 \rangle < \langle \lambda b_1 \rangle < \langle \lambda a_2 \rangle \quad \text{or} \quad \langle \lambda a_2 \rangle < \langle \lambda b_1 \rangle < \langle \lambda a_1 \rangle, \quad (\star)$$

where $\langle \cdot \rangle$ denotes the fractional part.

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where $\langle \cdot \rangle$ denotes the fractional part.

Theorem (Errera, 1913)

Condition $(\star)$ for all $1 \leq \lambda \leq N$ coprime to N is equivalent to Schwarz' classification.

Christol's Interlacing Criterion

[Christol 1986]: Study of **diagonals** of multivariate rational functions
(**Christol's conjecture**).

A by-product classification *à-la-Landau* of almost integral hypergeometric functions.

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Idea: Counting multiplicities of prime numbers p in

$$\frac{(a_1)_n \cdots (a_p)_n}{(b_1)_n \cdots (b_{p-1})_n n!}$$

amounts to the p -adic evaluation of elements of the arithmetic progressions $r_i + k \cdot N$ and $s_i + k \cdot N$, where $a_i = r_i/N$ and $b_i = s_i/N$.

Christol's Interlacing Criterion

Let $\langle \cdot \rangle : \mathbb{R} \rightarrow (0, 1]$ as the fractional part, where integers are assigned 1.
Define $\preceq$ on $\mathbb{R}^2$ via $a \preceq b$ if $\langle a \rangle < \langle b \rangle$ or $\langle a \rangle = \langle b \rangle$ and $a \geq b$.

Theorem (Christol, 1986)

Let $a_j, b_k \notin -\mathbb{N}$ be rational numbers and

$$F(x) := {}_pF_{p-1} \left[\begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_{p-1} \end{matrix}; x \right] = \sum_{n=0}^{\infty} \frac{(a_1)_n \cdots (a_p)_n}{(b_1)_n \cdots (b_{p-1})_n} \cdot \frac{x^n}{n!}.$$

Denote by N the least common denominator of a_j, b_k , and set $b_p = 1$. Then $F(x)$ is almost integral if and only if for all $1 \leq \lambda \leq N$ with $\gcd(\lambda, N) = 1$ we have for all $1 \leq k \leq p$ that

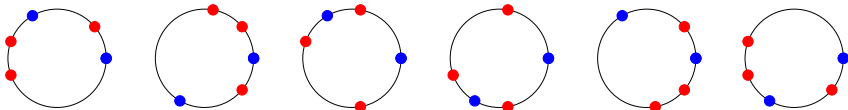
$$|\{\lambda a_j \preceq \lambda b_k : 1 \leq j \leq p\}| - |\{\lambda b_j \preceq \lambda b_k : 1 \leq j \leq p\}| \geq 0.$$

Christol's Interlacing Criterion graphically

Draw the sets $\{\exp(2\pi i \lambda a_j)\}$ in **red** and $\{\exp(2\pi i \lambda b_k)\}$ in **blue** on the unit circle for all $1 \leq \lambda \leq N$ with $\gcd(\lambda, N) = 1$. Then $F(x)$ is almost integral iff there are always at least as many **red** as **blue** points going counter-clockwise starting after 1.

Example

${}_3F_2([1/9, 4/9, 5/9], [1/3, 1]; x) = 1 + \frac{20}{3^5}x + \frac{2275}{3^{10}}x^2 + \frac{3124550}{3^{17}}x^3 + \dots$ is almost integral.



Beukers-Heckman Interlacing Criterion

Theorem (Christol 1986, Beukers-Heckman 1989, Katz 1990)

Let $a_j, b_k \notin -\mathbb{N}$ be rational numbers such that $a_j - b_k, a_j \notin \mathbb{Z}$ and

$$F(x) := {}_pF_{p-1} \left[\begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_{p-1} \end{matrix} ; x \right] = \sum_{n=0}^{\infty} \frac{(a_1)_n \cdots (a_p)_n}{(b_1)_n \cdots (b_{p-1})_n} \cdot \frac{x^n}{n!}.$$

Denote by N the least common denominator of a_j, b_k , and set $b_p = 1$. Then $F(x)$ is algebraic iff for all $1 \leq \lambda \leq N$ with $\gcd(\lambda, N) = 1$ we have for all $1 \leq k \leq p$ that

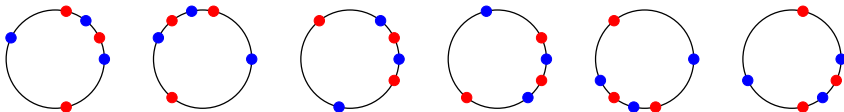
$$|\{\lambda a_j \preceq \lambda b_k : 1 \leq j \leq p\}| - |\{\lambda b_j \preceq \lambda b_k : 1 \leq j \leq p\}| = 0. \quad (\text{IC})$$

In other words, $F(x)$ is algebraic, if and only if the sets $\{2\pi i \lambda a_j\}$ and $\{2\pi i \lambda b_k\}$ *interlace* on the unit circle for all λ .

Beukers-Heckman Interlacing Criterion

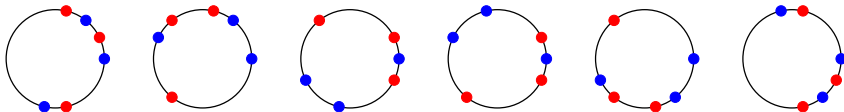
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$F(x) = {}_3F_2([1/14, 3/14, 11/14], [1/7, 3/7]; x)$ is algebraic:



Example

$F(x) = {}_3F_2([1/14, 3/14, 11/14], [1/7, 5/7]; x)$ is transcendental:



Beukers-Heckman vs Christol

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- [Beukers-Heckman 1989] has the additional assumption $a_j - b_k, a_j \notin \mathbb{Z}$.
- Assuming $a_j - b_k, a_j \notin \mathbb{Z}$, Christol's criterion is satisfied for $F(x)$ if and only if Beukers-Heckman interlacing holds.

PROPOSITION 3 : Toute fonction hypergéométrique F réduite et de hauteur 1 est globalement bornée si et seulement si, pour tout Δ tel que $(\Delta, N) = 1$, les nombres $\exp(2i\pi\Delta a_1)$ et $\exp(2i\pi\Delta b_1)$ sont entrelacés sur le cercle unité.

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- Assuming $a_j - b_k, a_j \notin \mathbb{Z}$, Christol's criterion is satisfied for $F(x)$ if and only if it is satisfied for **all** solutions of the hypergeometric differential equation.
- Under the ***p*-curvature conjecture** and $a_j - b_k, a_j \notin \mathbb{Z}$, the criteria are equivalent.

COROLLAIRE (modulo la conjecture de GROTHENDIECK si $s \geq 2$) : Une fonction hypergéométrique de hauteur 1 est globalement bornée si et seulement si elle est algébrique.

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COROLLAIRE (modulo la conjecture de GROTHENDIECK si $s > 2$) : Une fonction hypergéométrique de hauteur 1 est globalement bornée si et seulement si elle est algébrique.

- Hypergeometric equations are a special case of a class of equations for which the conjecture was solved by [Katz 1972].

Beukers-Heckman 1989

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- There is a Hermitian form invariant under the monodromy group, which is positive definite if and only if the parameters interlace on the unit circle.

Beukers-Heckman 1989

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- There is a Hermitian form invariant under the monodromy group, which is positive definite if and only if the parameters interlace on the unit circle.
- The monodromy group is discrete and contained in the unitary group, which is compact, if and only if interlacing holds.
- Complete classification of all possible resulting finite monodromy groups.

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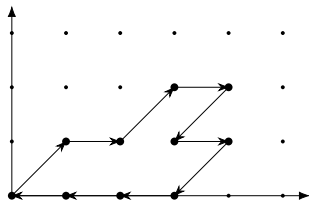
F. Beukers and G. Heckman

Table 8.3. (continued)

No.	Dimension	Parameter set		Field of definition	Group
2	3	$\frac{1}{14}, \frac{5}{14}, \frac{11}{14};$	$0, \frac{1}{2}, \frac{2}{3}$	$\mathbb{Q}(\sqrt{-7})$	ST 24
3			$0, \frac{1}{4}, \frac{3}{4}$		
4			$\frac{1}{3}, \frac{2}{3}, \frac{4}{3}$		
5	6	$0, \frac{1}{5}, \frac{4}{5};$	$\frac{1}{6}, \frac{1}{2}, \frac{5}{6}$	$\mathbb{Q}(\sqrt{5})$	ST 23
6			$\frac{1}{10}, \frac{1}{2}, \frac{9}{10}$		
7	8	$\frac{1}{6}, \frac{11}{30}, \frac{29}{30};$	$0, \frac{1}{3}, \frac{4}{5}$	$\mathbb{Q}(\omega, \sqrt{5})$	ST 27
8			$0, \frac{1}{3}, \frac{2}{3}$		

Example: Gessel Excursions

Gessel walks: Lattice walks in the quarter plane with step set $\{\rightarrow, \leftarrow, \nearrow, \swarrow\}$.



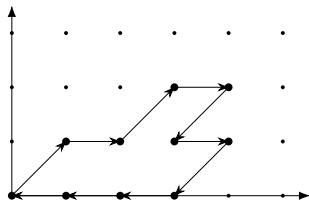
Consider the generating function

$$G(x) = \sum_{n \geq 0} g_n x^n$$

of **excursions** of length n .

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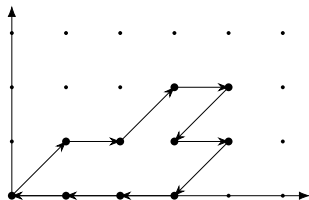
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Theorem (Kauers-Koutschan-Zeilberger, Bousquet-Mélou, Bostan-Kurkova-Raschel)

$$G(x) = \sum_{n \geq 0} \frac{(5/6)_n (1/2)_n}{(2)_n (5/3)_n} 16^n x^{2n} = {}_3F_2 \left[\begin{matrix} \frac{5}{6}, \frac{1}{2}, 1 \\ 2, \frac{5}{3} \end{matrix}; 16x^2 \right].$$

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Question: Is $G(x)$ algebraic?

Is $G(x) = {}_3F_2([5/6, 1/2, 1], [2, 5/3]; 16x^2)$ algebraic?

- Direct application of the interlacing criterion is not possible, as $a_3 = 1 \in \mathbb{Z}$.

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- **Trick:** Use identities for hypergeometric functions:

$$G(x) = \frac{1}{2x^2} \left({}_2F_1 \left[\begin{matrix} -1/2, -1/6 \\ 2/3 \end{matrix}; 16x^2 \right] - 1 \right),$$

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- Algebraicity of $G(x)$ was overlooked until [Bostan-Kauers 2010]; [Bostan 2017].
- The minimal polynomial $P(x, y)$ of $G(x)$ is

$$\begin{aligned} P(x, y) = & 27x^{14}y^8 + 108x^{12}y^7 + 189x^{10}y^6 + 189x^8y^5 - 9x^6(32x^4 + 28x^2 - 13)y^4 \\ & - 9x^4(64x^4 + 56x^2 - 5)y^3 - 2x^2(256x^6 - 312x^4 + 156x^2 - 5)y^2 \\ & - (32x^2 - 1)(4x^2 - 6x + 1)(4x^2 + 6x + 1)y - 256x^6 - 576x^4 + 48x^2 - 1 \end{aligned}$$

Example 2: Irrational Parameters

Consider the innocent recursion

$$u_{n+1} = \frac{2(2n+1)(n^2+2n-1)}{(n+1)(n^2-2)} u_n, \quad u_0 = 1.$$

The generating function

$$\sum_{n \geq 0} u_n x^n = {}_3F_2 \left[\begin{matrix} 1/2, \sqrt{2}+1, -\sqrt{2}+1 \\ \sqrt{2}, -\sqrt{2} \end{matrix} ; 4x \right] = \frac{(7x-1)(2x-1)}{(1-4x)^{5/2}}$$

is algebraic, but the criteria are not applicable.

Goal

The interlacing criteria treat the case of $a_j, b_k \in \mathbb{Q} \setminus \{-\mathbb{N}\}$
with $a_j - b_k, a_j \notin \mathbb{Z}$.

Aim: An easy-to-use algebraicity criterion to account for
irrational parameters and integer differences.

Examples

$${}_2F_1\left[\begin{matrix} 1, 1 \\ 2 \end{matrix}; x\right] = -\frac{\log(1-x)}{x}$$

$${}_2F_1\left[\begin{matrix} \frac{1}{2}, 1 \\ 2 \end{matrix}; 4x\right] = \frac{1-\sqrt{1-4x}}{2x}$$

$${}_3F_2\left[\begin{matrix} 1/2, \sqrt{2}+1, -\sqrt{2}+1 \\ \sqrt{2}, -\sqrt{2} \end{matrix}; 4x\right] = \frac{(7x-1)(2x-1)}{(1-4x)^{5/2}}$$

$${}_6F_5\left[\begin{matrix} \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 3, 3, 1 \\ \frac{1}{3}, \frac{2}{3}, 4, 2, 2 \end{matrix}; \frac{256}{27}x\right]$$

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$${}_3F_2\left[\begin{matrix} \frac{5}{6}, \frac{1}{2}, 1 \\ 2, \frac{5}{3} \end{matrix}; 16x^2\right]$$

Some Definitions

- Define

$$\mathcal{F}\left[\begin{matrix} c_1, \dots, c_r \\ d_1, \dots, d_s \end{matrix}; x\right] := \sum_{n \geq 0} \frac{(c_1)_n \cdots (c_r)_n}{(d_1)_n \cdots (d_s)_n} x^n.$$

- Note that

$${}_pF_q\left[\begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_q \end{matrix}; x\right] = \mathcal{F}\left[\begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_q, 1 \end{matrix}; x\right],$$
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- $F(x)$ is **contracted** if $c_j - d_k \notin \mathbb{N}$. $F(x)$ is **reduced** if $c_j - d_k \notin \mathbb{Z}$.

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- $F(x)$ is **contracted** if $c_j - d_k \notin \mathbb{N}$. $F(x)$ is **reduced** if $c_j - d_k \notin \mathbb{Z}$.
- The **contraction** $F^c(x)$ of $F(x)$ is obtained by removing pairs of parameters (c_j, d_k) with minimal difference $c_j - d_k \in \mathbb{N}$.
If $F(x)$ is given as ${}_pF_q$, convert to $\mathcal{F}$ first.

Examples

$${}_4F_3 \left[\begin{matrix} \frac{1}{3}, \frac{1}{2}, 2, 4 \\ \frac{3}{2}, 3, 1 \end{matrix}; x \right]^c = \mathcal{F} \left[\begin{matrix} \frac{1}{3}, \frac{1}{2}, 2, 4 \\ \frac{3}{2}, 3, 1, 1 \end{matrix}; x \right]^c = \mathcal{F} \left[\begin{matrix} \frac{1}{3}, \frac{1}{2} \\ \frac{3}{2}, 1 \end{matrix}; x \right] \quad \text{not reduced}$$

$${}_2F_1 \left[\begin{matrix} 1, 1 \\ 2 \end{matrix}; x \right]$$

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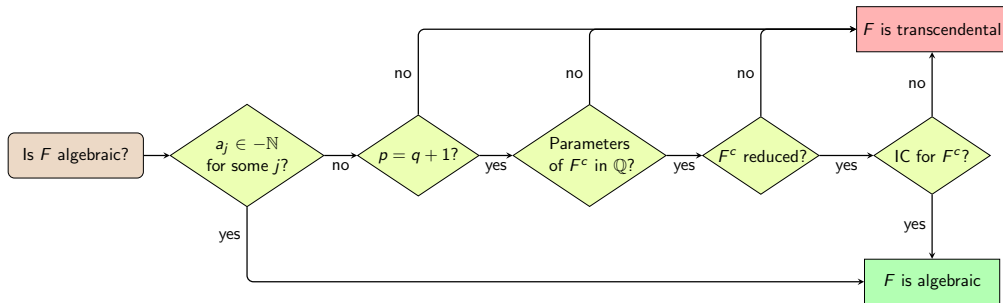
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The Criterion

Theorem (Fürnsinn-Y. 2024)

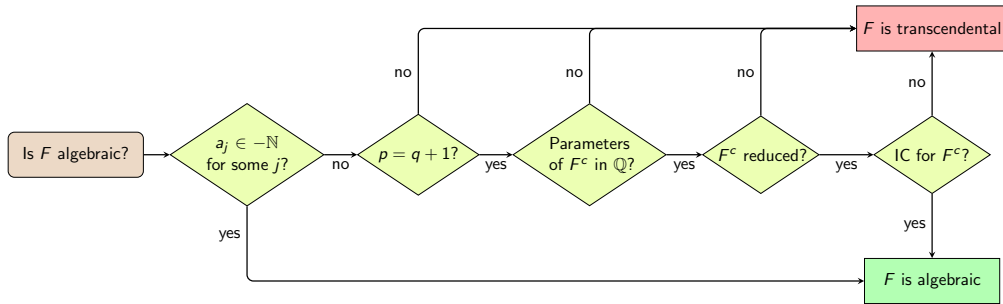
For any hypergeometric function $F(x) = {}_pF_q([a_1, \dots, a_p], [b_1, \dots, b_q]; x) \in \mathbb{Q}[[x]]$ the following decision tree answers the question whether it is algebraic.



[Caruso, Fürnsinn, 2026]: Implementation (many other algorithms).

Main steps of the proof

- Contraction preserves algebraicity.
- Algebraic contracted hypergeometric functions have rational parameters.
- Algebraic contracted hypergeometric functions are reduced.



Contraction Preserves Algebraicity

Proposition

The hypergeometric function $F(x)$ is algebraic if and only if $F^c(x)$ is algebraic.

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Proof sketch.

Contiguous relation:

$$(\theta + a_1)F(x) = a_1 \cdot {}_pF_{p-1} \left[\begin{matrix} a_1 + 1, a_2, \dots, a_p \\ b_1, b_2, \dots, b_{p-1} \end{matrix}; x \right].$$

Euclidean division of the hypergeometric operator $H(\theta)$ by $(\theta + \alpha)$:

$$H(\theta) = Q(\theta)(\theta + \alpha) + H(-\alpha), \quad H(-\alpha) = x \prod_{j=1}^p (-\alpha + a_j) + \alpha \prod_{k=1}^{p-1} (-\alpha + b_k - 1).$$

Thus, $0 = Q(\theta)(\theta + \alpha)F(x) + H(-\alpha)F(x)$. $H(-\alpha)$ vanishes iff $\alpha \in \{a_1, \dots, a_p\} \cap \{b_1 - 1, \dots, b_{p-1} - 1, 0\}$. Otherwise, $(\theta + \alpha)F(x)$ is algebraic iff $F(x)$ is algebraic. Iterating $\Rightarrow F$ algebraic iff F^c algebraic. □

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Alternative proof: [\[André 1989\]](#): $F(x)$ algebraic iff $F'(x)$ algebraic & almost integral.

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Lemma (Folklore, Singer 1980)

If $f(x)$ is algebraic then all solutions of $L_f^{\min}$ are algebraic.

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Lemma (Folklore, Singer 1980)

If $f(x)$ is algebraic then all solutions of $L_f^{\min}$ are algebraic.

Proposition

If $F(x)$ is contracted and has irrational parameters, it is transcendental.

Alternative proof: [Galočkin 1981] classification of hypergeometric **G-functions**.

Algebraic Contracted Hypergeometric Functions Are Reduced

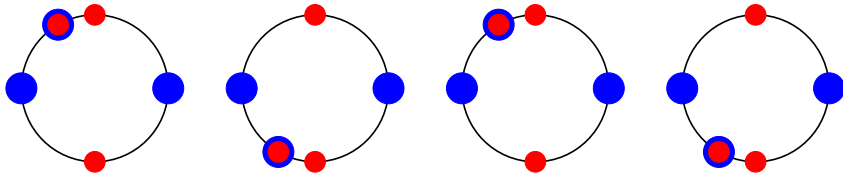
Proposition

If $F(x)$ is contracted but not reduced, then $F(x)$ is transcendental.

We prove that $F(x)$ is not almost integral using [\[Christol's criterion\]](#).

Proof "by example": Let

$$F(x) = {}_3F_2 \left[\begin{matrix} \frac{1}{4}, \frac{3}{4}, \frac{1}{3} \\ \frac{1}{2}, \frac{4}{3} \end{matrix}; x \right] \Rightarrow N = 12, \lambda \in \{1, 5, 7, 11\}$$



Recall: $\langle \cdot \rangle$ is the fractional part and $a \preceq b$ if $\langle a \rangle < \langle b \rangle$ or $\langle a \rangle = \langle b \rangle$ and $a \geq b$.

Example 1

Let:
$$u_{n+1} = \frac{(14n+1)(14n+3)(14n+11)(n^2+2n+4)}{56(7n+1)(7n+3)(n+3)(n^2+3)} u_n, \quad u_0 = 1$$

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Generating function:

$$f(x) = \mathcal{F} \left[\begin{matrix} \frac{1}{14}, \frac{3}{14}, \frac{11}{14}, 1+i\sqrt{3}, 1-i\sqrt{3} \\ \frac{1}{7}, \frac{3}{7}, i\sqrt{3}, -i\sqrt{3}, 3 \end{matrix} ; x \right] = {}_6F_5 \left[\begin{matrix} \frac{1}{14}, \frac{3}{14}, \frac{11}{14}, 1+i\sqrt{3}, 1-i\sqrt{3}, 1 \\ \frac{1}{7}, \frac{3}{7}, i\sqrt{3}, -i\sqrt{3}, 3 \end{matrix} ; x \right].$$

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Contraction has rational parameters and is reduced:

$$f^c(x) = \mathcal{F} \left[\begin{matrix} \frac{1}{14}, \frac{3}{14}, \frac{11}{14} \\ \frac{1}{7}, \frac{3}{7}, 3 \end{matrix} ; x \right] = {}_4F_3 \left[\begin{matrix} \frac{1}{14}, \frac{3}{14}, \frac{11}{14}, 1 \\ \frac{1}{7}, \frac{3}{7}, 3 \end{matrix} ; x \right].$$

Example 1

$$\text{Let: } u_{n+1} = \frac{(14n+1)(14n+3)(14n+11)(n^2+2n+4)}{56(7n+1)(7n+3)(n+3)(n^2+3)} u_n, \quad u_0 = 1$$

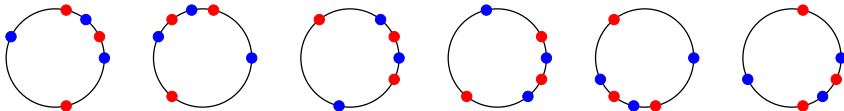
Generating function:

$$f(x) = \mathcal{F} \left[\begin{matrix} \frac{1}{14}, \frac{3}{14}, \frac{11}{14}, 1+i\sqrt{3}, 1-i\sqrt{3} \\ \frac{1}{7}, \frac{3}{7}, i\sqrt{3}, -i\sqrt{3}, 3 \end{matrix} ; x \right] = {}_6F_5 \left[\begin{matrix} \frac{1}{14}, \frac{3}{14}, \frac{11}{14}, 1+i\sqrt{3}, 1-i\sqrt{3}, 1 \\ \frac{1}{7}, \frac{3}{7}, i\sqrt{3}, -i\sqrt{3}, 3 \end{matrix} ; x \right].$$

Contraction has rational parameters and is reduced:

$$f^c(x) = \mathcal{F} \left[\begin{matrix} \frac{1}{14}, \frac{3}{14}, \frac{11}{14} \\ \frac{1}{7}, \frac{3}{7}, 3 \end{matrix} ; x \right] = {}_4F_3 \left[\begin{matrix} \frac{1}{14}, \frac{3}{14}, \frac{11}{14}, 1 \\ \frac{1}{7}, \frac{3}{7}, 3 \end{matrix} ; x \right].$$

$f^c(x)$ is **algebraic** by the interlacing criterion, thus so is $f(x)$.



Example 2

$$u_n = \frac{3}{2} \binom{4n}{n} \frac{n+2}{(n+1)(n+3)}.$$

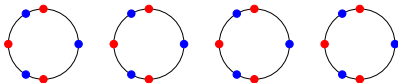
Generating function:

$$f(x) = {}_6F_5 \left[\begin{matrix} \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 3, 3, 1 \\ \frac{1}{3}, \frac{2}{3}, 4, 2, 2 \end{matrix} ; \frac{256}{27}x \right]$$

Contraction:

$$f^c(x) = {}_4F_3 \left[\begin{matrix} \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1 \\ \frac{1}{3}, \frac{2}{3}, 4 \end{matrix} ; \frac{256}{27}x \right]$$

Interlacing criterion: $f(x)$ algebraic.



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$$u_n = \frac{3}{2} \binom{4n}{n} \frac{n+2}{(n+1)(n+3)}.$$

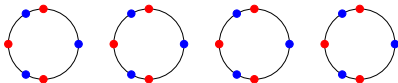
Generating function:

$$f(x) = {}_6F_5 \left[\begin{matrix} \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 3, 3, 1 \\ \frac{1}{3}, \frac{2}{3}, 4, 2, 2 \end{matrix} ; \frac{256}{27}x \right]$$

Contraction:

$$f^c(x) = {}_4F_3 \left[\begin{matrix} \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1 \\ \frac{1}{3}, \frac{2}{3}, 4 \end{matrix} ; \frac{256}{27}x \right]$$

Interlacing criterion: $f(x)$ algebraic.



$$v_n = \frac{3}{2} \binom{4n}{n} \frac{n+2}{(n+1)^2}.$$

Generating function:

$$g(x) = {}_6F_5 \left[\begin{matrix} \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 3, 1, 1 \\ \frac{1}{3}, \frac{2}{3}, 2, 2, 2 \end{matrix} ; \frac{256}{27}x \right]$$

Contraction:

$$g^c(x) = {}_5F_4 \left[\begin{matrix} \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1, 1 \\ \frac{1}{3}, \frac{2}{3}, 2, 2 \end{matrix} ; \frac{256}{27}x \right]$$

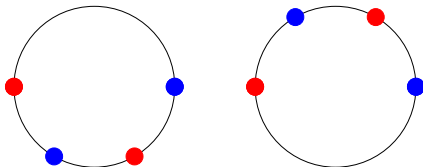
Not reduced: $g(x)$ not algebraic.

Example 3 – Gessel Revisited

Recall the generating function of Gessel excursions

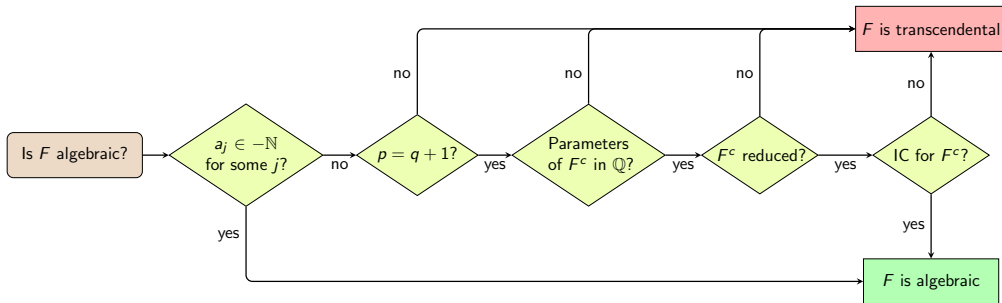
$$G(x) = {}_3F_2\left[\begin{matrix} \frac{5}{6}, \frac{1}{2}, 1 \\ 2, \frac{5}{3} \end{matrix}; 16x^2\right] = \mathcal{F}\left[\begin{matrix} \frac{5}{6}, \frac{1}{2} \\ 2, \frac{5}{3} \end{matrix}; x\right].$$

$G(x)$ is contracted, reduced, has rational parameters, satisfies the interlacing criterion:



The End

Thank you for your attention!



Bonus: Algebraicity of $F(x)$ vs $F'(x)$

Corollary

For a hypergeometric function $F(x) \in \mathbb{Q}[[x]]$,

$$F(x) \text{ is algebraic} \iff F'(x) \text{ is algebraic.}$$

Bonus: Algebraicity of $F(x)$ vs $F'(x)$

Corollary

For a hypergeometric function $F(x) \in \mathbb{Q}[[x]]$,

$$F(x) \text{ is algebraic} \iff F'(x) \text{ is algebraic.}$$

$\log(1-x)$ is not hypergeometric and the derivative of $\log(1-x)/x$ is not algebraic.

Proof.

“ $\Rightarrow$ ”: Derivatives of algebraic series are algebraic.

“ $\Leftarrow$ ”: Use the identity

$$\frac{d}{dx} {}_pF_q \left[\begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_q \end{matrix}; x \right] = \frac{a_1 \cdots a_p}{b_1 \cdots b_q} \cdot {}_pF_q \left[\begin{matrix} a_1 + 1, \dots, a_p + 1 \\ b_1 + 1, \dots, b_q + 1 \end{matrix}; x \right].$$



Bonus: Galočkin's Theorem on Irrational Parameters

Theorem (Galočkin 1981)

A hypergeometric function ${}_pF_q([a_1, \dots, a_p], [b_1, \dots, b_q]; x)$ is a **G-function** if and only if its irrational parameters can be arranged in pairs (a_{j_ν}, b_{k_ν}) with $a_{j_\nu} - b_{k_\nu} \in \mathbb{N}$.

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In other words: A hypergeometric function is a G-function iff its contraction has only rational parameters.